

Part 02

1. What we mean by Generalization concept using Generics?

- **Generalization using Generics** means writing reusable code that can work with **different data types** without duplicating the same code for each type.
- Example: `List<T>` can store `int`, `string`, or any other type using the same generic class.

2. What we mean by hierarchy design in real business?

Hierarchy Design in Real Business

Hierarchy design means organizing related classes or entities into **levels based on their common characteristics and relationships**, usually using **inheritance**.

For example, in a company:

```
Employee
├─ Manager
├─ Developer
└─ Accountant
```

`Employee` contains common properties such as:

```
</> C#
Name
Id
Salary
```

while each derived class can have its own specific behavior.

```
</> C#

class Employee
{
    public string Name { get; set; }
    public decimal Salary { get; set; }
}

class Manager : Employee
{
    public int TeamSize { get; set; }
}

class Developer : Employee
{
    public string ProgrammingLanguage { get; set; }
}
```

Why is this useful in real business?

It allows us to:

- **Avoid code duplication** by putting common features in the base class.
- **Represent real-world relationships** clearly.
- **Add specialized behavior** to different types.
- **Use polymorphism** to treat different employees as `Employee` when appropriate.
- **Make the system easier to maintain and extend.**